

Credit Name: CSE2130 - File Structures & Exception Handling
Assignment Name: StudentSemesterAverage

Understanding the Problem

How did you approach understanding the challenge?

Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?

I understood the problem going in. The only thing I wasn't certain on was using the GUI, but I knew I could refresh my memory by looking at my previous projects.

Planning the Solution

Did you create a plan or break the problem into smaller steps before coding?

How did you decide on the tools, data structures, or algorithms to use?

My plan was very general. I planned on first setting up the GUI using WindowBuilder. From there, I would add and import the necessary file structure objects. I figured that the logic wasn't too complicated, so I felt confident making things up as I went. The only main concern was creating that notification window every time you saved the inputted information to the file, but I thought it wouldn't be too difficult.

Implementation

Did you write the code in small pieces or attempt the entire solution at once?

How did you test your solution along the way to make sure it was working?

I started by creating the foundation of the program using WindowBuilder. After all of the elements were created and named, I started by adding an action event to the "Save to File" button. In it, I created all the variables and objects I would need to calculate and write the information that would eventually be saved in the file. I then worked on the "View File Contents" button. All that was needed was the code to read and display the information from the file. Once that was completed, I coded the little confirmation window that pops up every time you save the information. I finished by addressing some minor issues that occurred.

Overcoming Challenges

What part of the problem was the most difficult for you?

How did you handle moments when you felt unsure of what to do next?

There were three major issues that I encountered while coding the project. The first was that I had no idea how I was supposed to make the confirmation window

appear. This wasn't too difficult to deal with, as a couple quick searches online gave me the answers I needed. The second problem was that the `writeObject()` method I was originally using overwrote the text in the file every time I used, rather than just adding to it. I attempted to search up some answers on Google, but I struggled to find a concrete solution. I ended up asking my teacher for help. He told me to use the simpler `write()` method instead, and that ended up working well. The last issue was very similar. I was using the `setText()` method to display the information in the file. However, because I was calling the method for each line in the file, it was overwriting the information on the text area. To fix this, I found a method online called `.append` which would add to the text area rather than removing the initial information and writing new text.

Learning

Was there anything you learned that you think will help you with future challenges?

Most of the methods and logic I used for this project came from previous projects, including the GUI part. However, I did learn how to make a pop up window, and I also learned about new methods that may be helpful in the future.